SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

CO5_AT1 — UNCERTAINTY VISUALIZATION LAB

Course Code:	DSA06	**Course Title:**	Data Handling and Visualization
CO Assessed:	CO5 – Analysis (BL4)	**Assessment:**	Assessment Tool 1 – Uncertainty Visualization Lab
Weightage:	20% of CIA	**Total Marks:**	25 (5 Questions × 5 Marks)
CO5 Statement:	*Analyze and construct visualizations that communicate uncertainty using frequency framing, confidence intervals, trend uncertainty bands, HOPs, bootstrapping, and appropriate axis scales.(BL4)*		
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Section / Batch:		**Date:**	27.08.2026

Code of Conduct

I Priyanka.G / 192524124 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: ___________________________

Instructions

- This test contains 5 questions, each carrying 5 marks. Total: 25 Marks.
- No negative marking. Unanswered questions carry zero marks.
- No calculators or electronic devices permitted.
- Duration: 90 minutes.
- Show all supporting calculations (e.g., confidence interval bounds, residual standard deviation, bootstrap resample means) where relevant.
- Every diagram must include a title, correctly labeled axes, and a legend where relevant.
- Diagrams may be hand-drawn or clearly described in words if drawing tools are unavailable.

Annexure A – UNCERTAINTY VISUALIZATION LAB

Faculty In-charge	Course Coordinator	Head of Department
Signature: ________________	Signature: ________________	Signature: ________________
Date: ____________________	Date: ____________________	Date: ____________________

Problem 46 - Hypothetical Outcome Plot - Server Response Time (5 Marks)

Data: A server's response time has a true mean of 120 ms and a standard deviation of 10 ms.

***Task:* List 5 plausible individual draws that could each form one frame of a Hypothetical Outcome Plot, and explain what a viewer learns from watching all 5 frames in sequence versus a single error bar.**

Analysis Plan (Key Elements):

Technique	Hypothetical Outcome Plot (HOP) - sequential frames of plausible individual draws
Distribution	Assumed approximately Normal: mean = 120 ms, SD = 10 ms
Comparison	A single static error bar vs. watching 5 individual outcome frames in sequence

Construction & Analysis:

Five plausible individual draws from this distribution (mean = 120 ms, SD = 10 ms), matching the constructed HOP frames (Figure 46), are: 112 ms, 125 ms, 118 ms, 132 ms, and 121 ms. Each of these values is well within a normal, expected range for this distribution: 112 ms is about 0.8 SD below the mean; 125 ms is about 0.5 SD above; 118 ms is about 0.2 SD below; 132 ms is about 1.2 SD above; and 121 ms is about 0.1 SD above - all comfortably inside the range a normal distribution with SD = 10 would be expected to produce on a typical draw.

A single error bar (for example, mean ± 1 SD, drawn as 120 ± 10 ms, or a 95% confidence interval) only ever shows a static, aggregate summary of the uncertainty: it communicates the plausible range for the parameter or distribution as a whole, but it does not show what any one concrete, individual response time might actually look like, nor does it convey how much a real measurement can bounce around from one request to the next. A viewer reading a static error bar has to mentally translate an abstract range into 'what could a single response actually be,' a step research on uncertainty visualization has repeatedly found people do poorly - static intervals are frequently misread as showing a fixed, certain boundary rather than a summary of variability across many possible draws.

Watching all 5 frames of the HOP in sequence instead lets a viewer directly perceive individual, concrete outcomes one at a time - seeing the response time actually land at 112 ms in one frame, then jump to 125 ms in the next, then 118 ms, then spike to 132 ms, then settle at 121 ms - which builds a felt, experiential sense of how much any single request's response time can vary, in a way a static bar cannot. This is precisely the design motivation behind HOPs: by animating through individually plausible draws rather than only summarizing them into one range, a HOP corrects the common misreading of uncertainty visualizations as showing fixed, certain outcomes, replacing it with a direct demonstration of the sampling variability itself.

Constructed Chart:

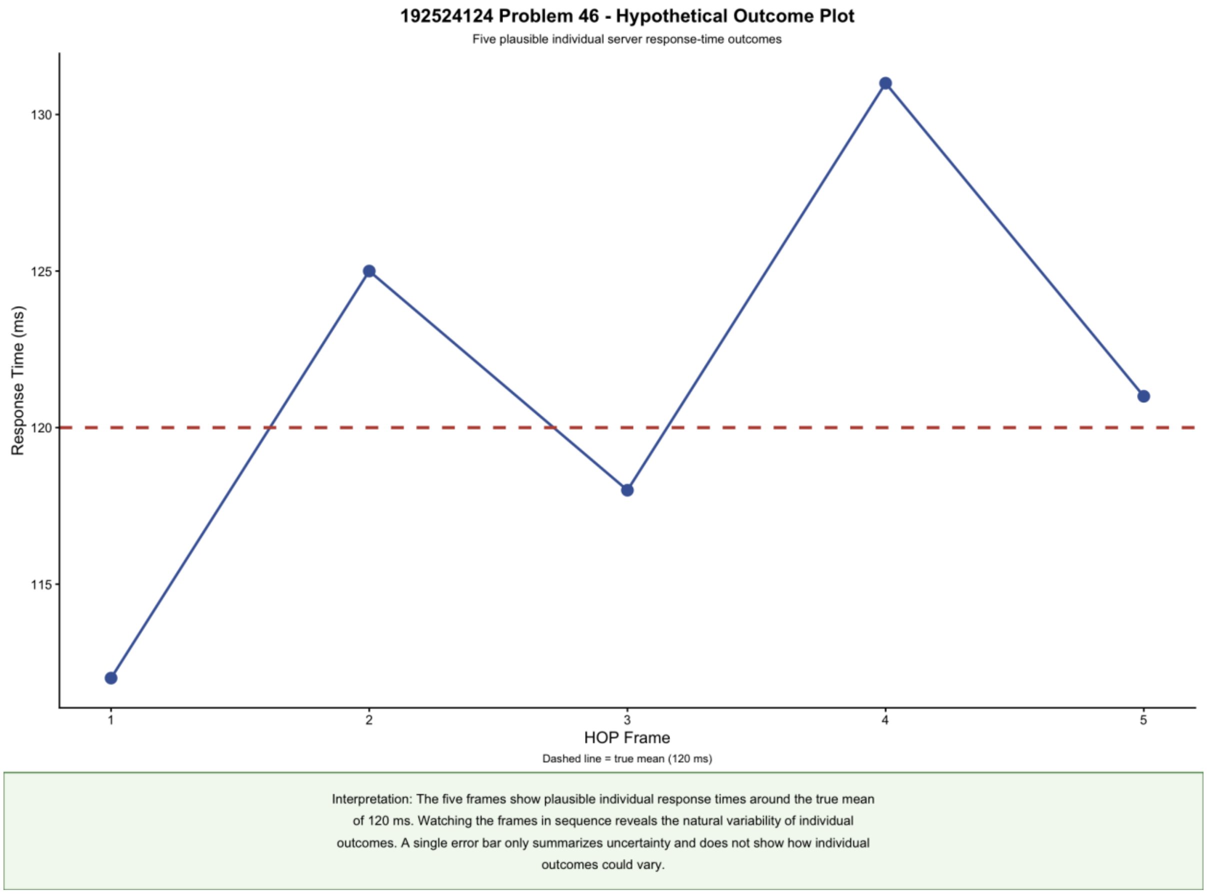

Figure 46: Hypothetical Outcome Plot - Five Plausible Server Response-Time Draws

Problem 47 - Bootstrap Resampling - Machine Output Counts (5 Marks)

Given Data:

Observation	1	2	3	4	5
Output Count	64	66	74	77	77

***Task:* Perform one bootstrap resample (draw 5 values with replacement) and calculate its mean, then explain how repeating this process thousands of times and taking the middle 95% of resulting means gives a confidence interval.**

Analysis Plan (Key Elements):

Technique	Bootstrap resampling with replacement, n = 5
Original Sample Mean	(64 + 66 + 74 + 77 + 77) ÷ 5 = 71.6
One Resample Given	(77, 64, 77, 74, 77) → mean = 73.8
Purpose	Build a percentile bootstrap confidence interval without assuming a population distribution shape

Construction & Analysis:

A bootstrap resample is constructed by drawing 5 values from the original 5 observations (64, 66, 74, 77, 77) with replacement - meaning each draw is made independently from the full original set, so the same value can be selected more than once, and some original values may not be selected at all in a given resample.

One such resample (matching the constructed chart, Figure 47) is (77, 64, 77, 74, 77). This resample happens to include the value 77 three times - legitimate, since 77 appears in the original data and each draw is independent - while the value 66 was not selected at all in this particular resample, directly illustrating the 'with replacement' mechanism in action.

The mean of this resample is calculated as: (77 + 64 + 77 + 74 + 77) ÷ 5 = 369 ÷ 5 = 73.8. For comparison, the original sample's own mean is (64 + 66 + 74 + 77 + 77) ÷ 5 = 358 ÷ 5 = 71.6 - this single resample's mean (73.8) is noticeably higher than the original sample mean (71.6), which is expected and illustrates that any one resample's mean is itself a somewhat noisy estimate, shifted by which values happened to be drawn (and drawn multiple times, in the case of 77) that particular time.

To build a full confidence interval, this resampling process is repeated thousands of times (for example, 10,000 times): each repetition draws a fresh set of 5 values with replacement from the same original 5 observations and records that resample's mean. This produces a large distribution of bootstrap means that approximates the true sampling distribution of the mean, without requiring any assumption about the shape of the underlying population (unlike a classical formula-based confidence interval, which typically assumes normality).

To construct a 95% percentile bootstrap confidence interval from this distribution of thousands of bootstrap means, the means are sorted from smallest to largest, and the value at the 2.5th percentile is taken as the lower bound while the value at the 97.5th percentile is taken as the upper bound - together capturing the middle 95% of all recorded bootstrap means. This interval is interpreted as the range within which the true population mean output count is estimated to lie with 95% confidence, built entirely empirically by resampling the observed data itself rather than relying on a parametric formula.

Constructed Chart:

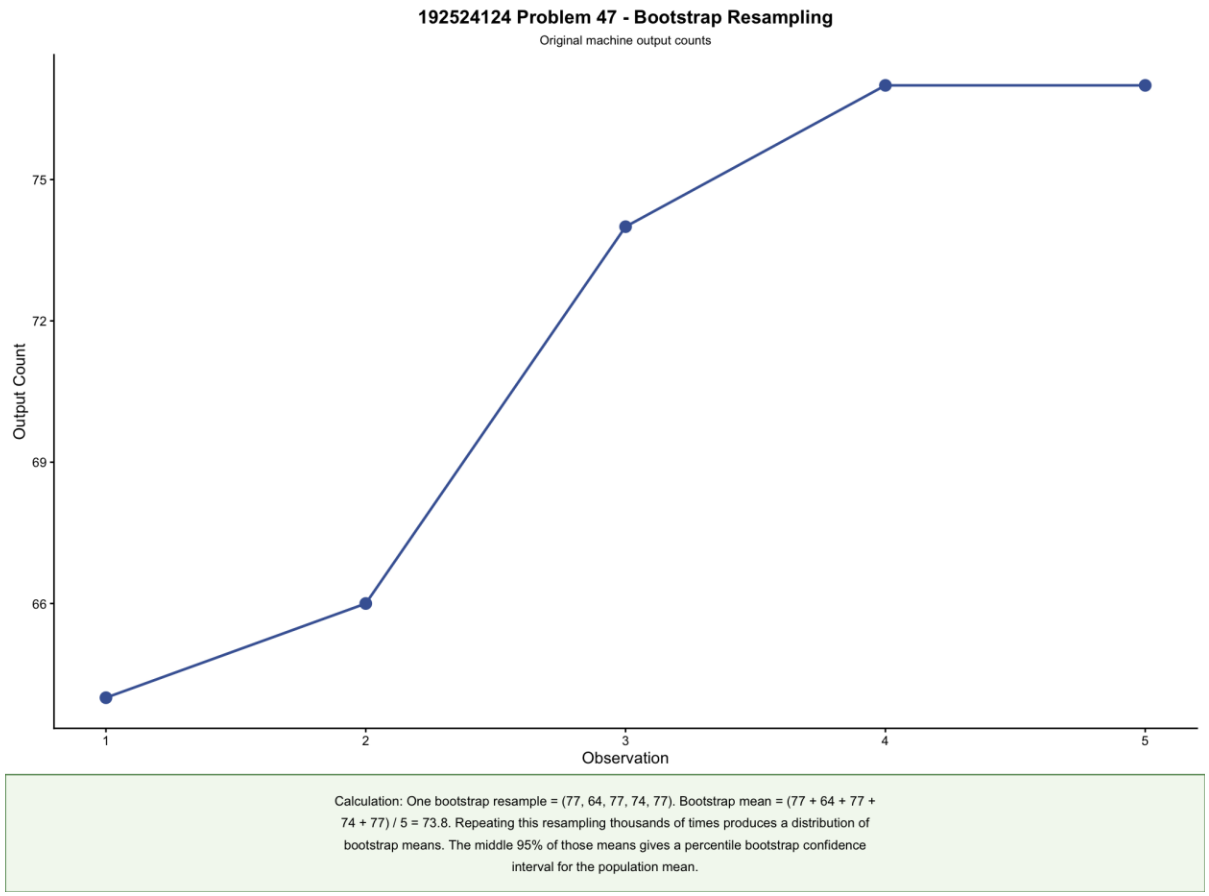

Figure 47: Bootstrap Resampling - Original Machine Output Counts and Resample Calculation

Problem 48 - Linear vs. Logarithmic Axis - Social Media Follower Counts (5 Marks)

Given Data:

Entity	Entity 1	Entity 2	Entity 3	Entity 4
Followers	80	2,500	60,000	1,800,000

***Task:* Plot this data on both a linear and a logarithmic axis, calculate the ratio between each consecutive pair of values, and explain which axis reveals the data's structure more clearly, and why.**

Analysis Plan (Key Elements):

Chart Type	Dual-panel bar chart - same data on a linear axis and a logarithmic axis
Consecutive Ratios	31.25× (Entity 1→2), 24× (Entity 2→3), 30× (Entity 3→4)
Data Structure	Roughly constant multiplicative growth (exponential-type pattern), not additive growth
Conclusion	Logarithmic axis reveals the structure far more clearly than the linear axis

Construction & Analysis:

The ratio between each consecutive pair of follower counts is calculated as: Entity 2 ÷ Entity 1 = 2,500 ÷ 80 = 31.25×; Entity 3 ÷ Entity 2 = 60,000 ÷ 2,500 = 24×; Entity 4 ÷ Entity 3 = 1,800,000 ÷ 60,000 = 30×. All three ratios cluster tightly in the 24×-31.25× range, meaning this data grows at a roughly constant multiplicative rate from one entity to the next - each entity has, very approximately, 24 to 31 times as many followers as the previous one - rather than growing by a constant additive amount. This is the signature of exponential-type (multiplicative) growth, not linear growth.

On the linear axis (left panel of Figure 48), the largest value (1,800,000) is nearly 22,500 times the smallest (80). An axis scaled to fit 1,800,000 has essentially no visual resolution left for the smaller values: both Entity 1 (80) and Entity 2 (2,500) round visually to bars of imperceptible height, indistinguishable from the axis baseline, even though Entity 2 genuinely has over 31 times as many followers as Entity 1. The linear axis can only 'afford' enough vertical resolution to display the single largest value clearly, crushing every smaller value into invisibility.

On the logarithmic axis (right panel of Figure 48), equal visual distances represent equal multiplicative steps rather than equal additive amounts, so a 24× jump and a 31.25× jump - despite being different absolute magnitudes - occupy visually similar-sized steps up the axis, since $\log_{10}(24) \approx 1.38$ and $\log_{10}(31.25) \approx 1.49$ are close in value. This lets all four bars remain clearly visible and individually comparable on the log axis, with their heights directly reflecting the consistent order-of-magnitude growth pattern present in the underlying ratios.

The logarithmic axis therefore reveals the data's actual structure far more clearly than the linear axis. Because the data itself grows multiplicatively (with roughly constant ratios around 24×-31×) rather than additively, and a logarithmic scale is specifically designed to make constant-ratio growth appear as an even, visually comparable step pattern, the log axis is the correct choice for this dataset - the linear axis's apparent 'structure' (one enormous bar, three invisible ones) is an artefact of the scale choice, not a true reflection of the underlying data.

Constructed Chart:

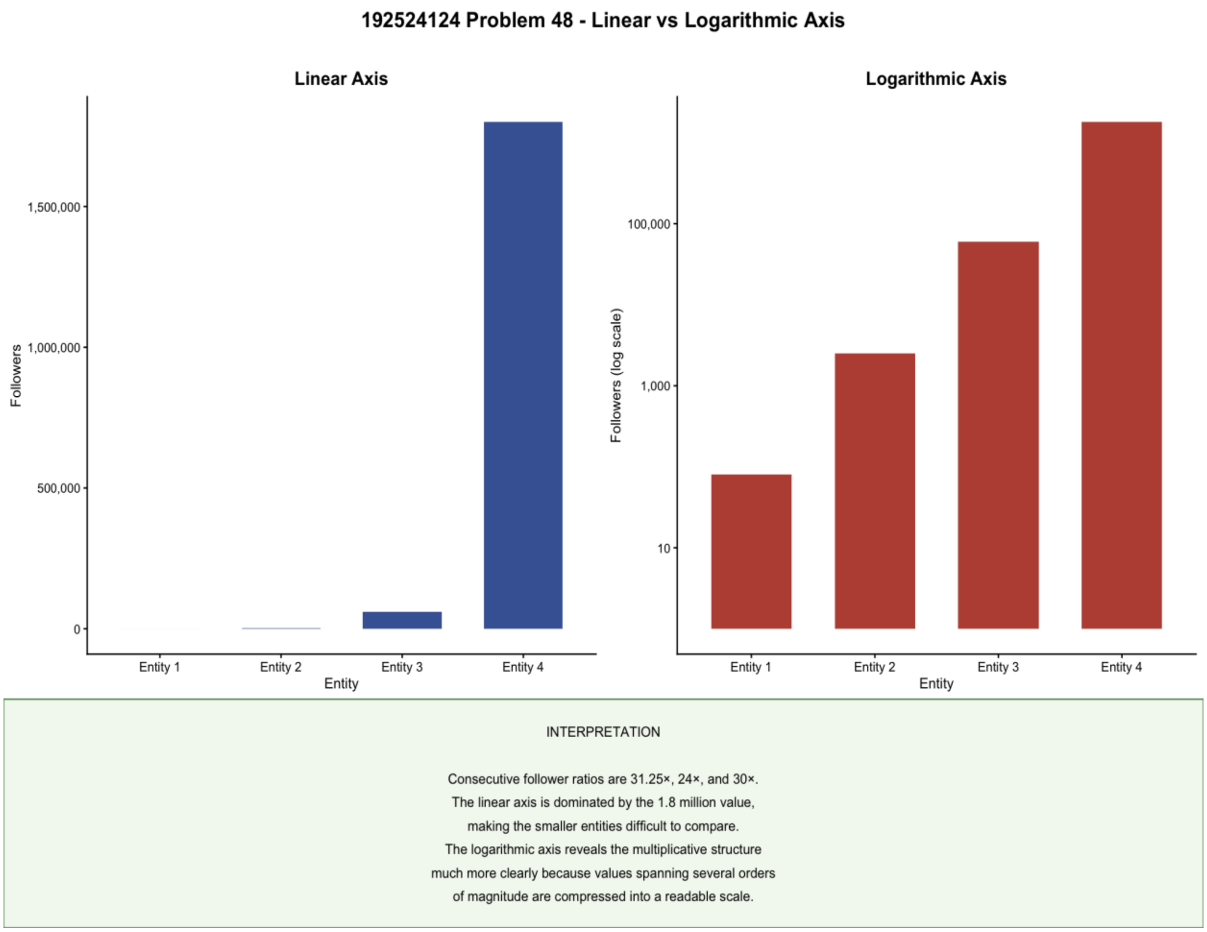

Figure 48: Follower Counts - Linear Axis (left) vs. Logarithmic Axis (right)

Problem 49 - Bootstrap Resampling - Survey Satisfaction Scores (5 Marks)

Given Data:

Observation	1	2	3	4	5
Satisfaction Score	33	36	39	44	46

Task: ***Perform one bootstrap resample (draw 5 values with replacement) and calculate its mean, then explain how repeating this process thousands of times and taking the middle 95% of resulting means gives a confidence interval.***

Analysis Plan (Key Elements):

Technique	Bootstrap resampling with replacement, n = 5
Original Sample Mean	(33 + 36 + 39 + 44 + 46) ÷ 5 = 39.6
One Resample Given	(44, 46, 39, 46, 33) → mean = 41.6
Purpose	Build a percentile bootstrap confidence interval for the population mean satisfaction score

Construction & Analysis:

A bootstrap resample is constructed by drawing 5 values from the original 5 satisfaction scores (33, 36, 39, 44, 46) with replacement, so the same score can be selected more than once and some original scores may be left out entirely in a given resample.

One such resample (matching the constructed chart, Figure 49) is (44, 46, 39, 46, 33). Here the value 46 has been drawn twice - a legitimate outcome of independent sampling with replacement - while the value 36 was not drawn at all in this particular resample.

The mean of this resample is calculated as: (44 + 46 + 39 + 46 + 33) ÷ 5 = 208 ÷ 5 = 41.6. The original sample's own mean is (33 + 36 + 39 + 44 + 46) ÷ 5 = 198 ÷ 5 = 39.6 - this resample's mean (41.6) is somewhat higher than the original sample mean (39.6), again illustrating that a single resample's mean is a noisy estimate that shifts depending on exactly which scores were drawn, and how many times, in that particular resample.

To build a full confidence interval for the population's true mean satisfaction score, this resampling process is repeated thousands of times, each time drawing a fresh set of 5 values with replacement from the same original 5 scores and recording that resample's mean. This builds up a large empirical distribution of bootstrap means that approximates the true sampling distribution of the mean, without assuming the underlying population of satisfaction scores follows any particular shape.

A 95% percentile bootstrap confidence interval is then formed by sorting all the recorded bootstrap means from smallest to largest and taking the 2.5th percentile as the lower bound and the 97.5th percentile as the upper bound, capturing the middle 95% of the bootstrap means. This gives a data-driven estimate of the range within which the true population mean satisfaction score most plausibly lies, built entirely from resampling the 5 observed scores rather than from a parametric formula that would require assuming, for example, that satisfaction scores are normally distributed.

Constructed Chart:

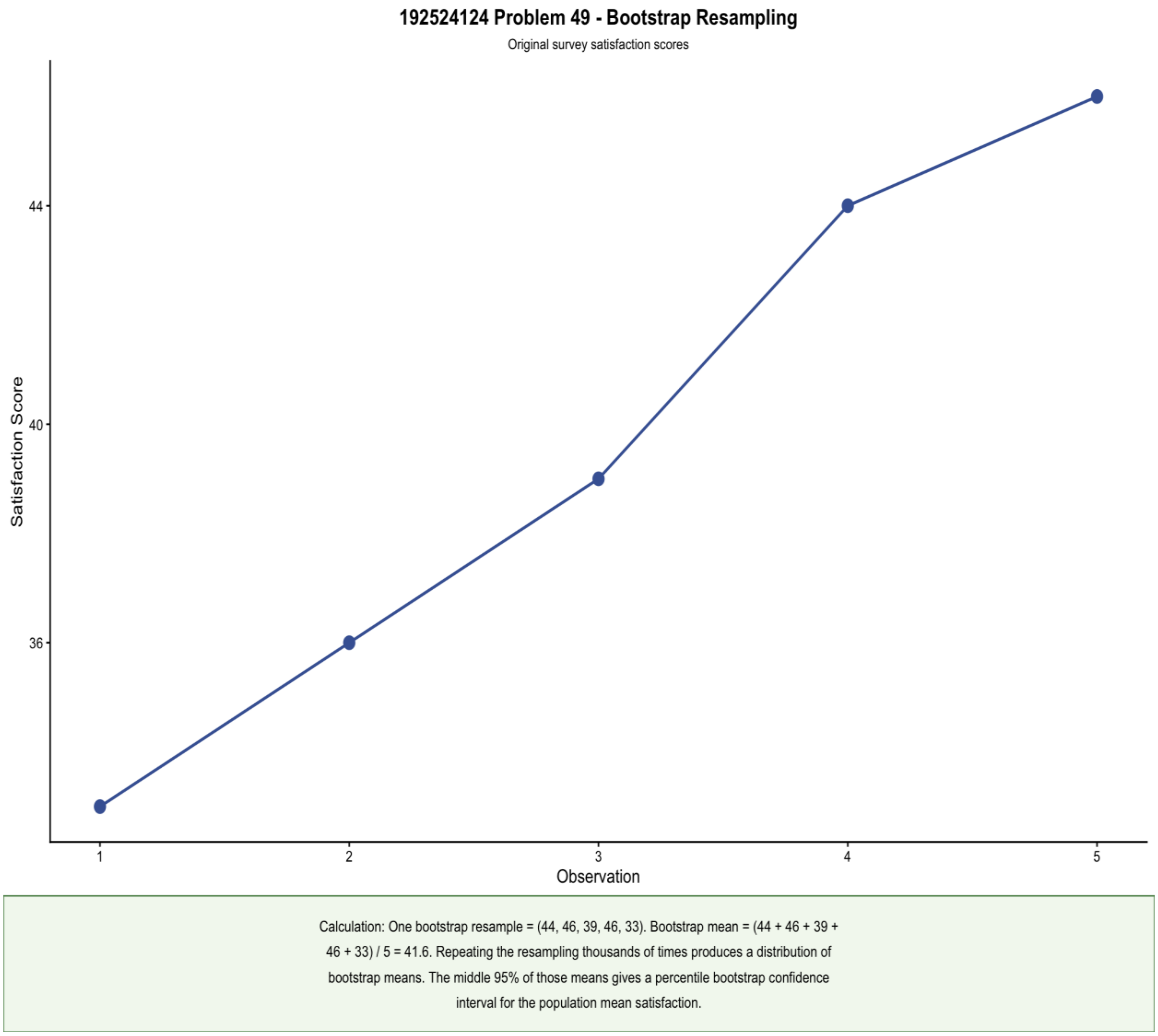

Figure 49: Bootstrap Resampling - Original Survey Satisfaction Scores and Resample Calculation

Problem 50 - Linear vs. Logarithmic Axis - National GDP Figures (5 Marks)

Given Data:

Entity	Entity 1	Entity 2	Entity 3	Entity 4
GDP (₹ billion)	2	40	900	21,000

Task: ***Plot this data on both a linear and a logarithmic axis, calculate the ratio between each consecutive pair of values, and explain which axis reveals the data's structure more clearly, and why.***

Analysis Plan (Key Elements):

Chart Type	Dual-panel bar chart - same data on a linear axis and a logarithmic axis
Consecutive Ratios	20× (Entity 1→2), 22.5× (Entity 2→3), 23.33× (Entity 3→4)
Data Structure	Roughly constant multiplicative growth, not additive growth
Conclusion	Logarithmic axis reveals the structure far more clearly than the linear axis

Construction & Analysis:

The ratio between each consecutive pair of GDP figures is calculated as: Entity 2 ÷ Entity 1 = 40 ÷ 2 = 20×; Entity 3 ÷ Entity 2 = 900 ÷ 40 = 22.5×; Entity 4 ÷ Entity 3 = 21,000 ÷ 900 ≈ 23.33×. All three ratios sit tightly in the 20×-23.33× range, meaning national GDP grows at a roughly constant multiplicative rate across these four entities - each economy is, very approximately, 20 to 23 times larger than the previous one - the same constant-ratio (multiplicative) growth signature seen in Problem 48, rather than a constant additive step between entities.

On the linear axis (left panel of Figure 50), the largest GDP (₹21,000 billion) is 10,500 times the smallest (₹2 billion). With the axis scaled to accommodate ₹21,000 billion, Entity 1 (₹2 billion) and Entity 2 (₹40 billion) both round to visually imperceptible bar heights indistinguishable from the baseline, despite Entity 2 genuinely being 20 times larger than Entity 1 - the linear axis has essentially no resolution left to show that real difference once it must also accommodate the far larger Entity 4.

On the logarithmic axis (right panel of Figure 50), equal visual distances represent equal multiplicative steps, so the 20×, 22.5×, and 23.33× jumps between successive entities - despite spanning vastly different absolute magnitudes (₹2 billion to ₹40 billion versus ₹900 billion to ₹21,000 billion) - occupy visually similar-sized steps up the axis, since $\log_{10}(20) \approx 1.30$, $\log_{10}(22.5) \approx 1.35$, and $\log_{10}(23.33) \approx 1.37$ are all close in value. This keeps all four bars clearly visible and directly comparable, with their heights reflecting the genuinely consistent growth-rate pattern in the underlying ratios.

As with the follower-count data in Problem 48, the logarithmic axis reveals this GDP data's actual structure far more clearly than the linear axis. Because the underlying economic growth pattern here is multiplicative (consistently around 20×-23× per step) rather than additive, and because a log scale is purpose-built to display constant-ratio growth as an even, comparable pattern, the log axis is the more informative and accurate choice - the linear axis's single towering bar and three invisible ones misrepresent the data's true, evenly-scaled multiplicative structure.

Constructed Chart:

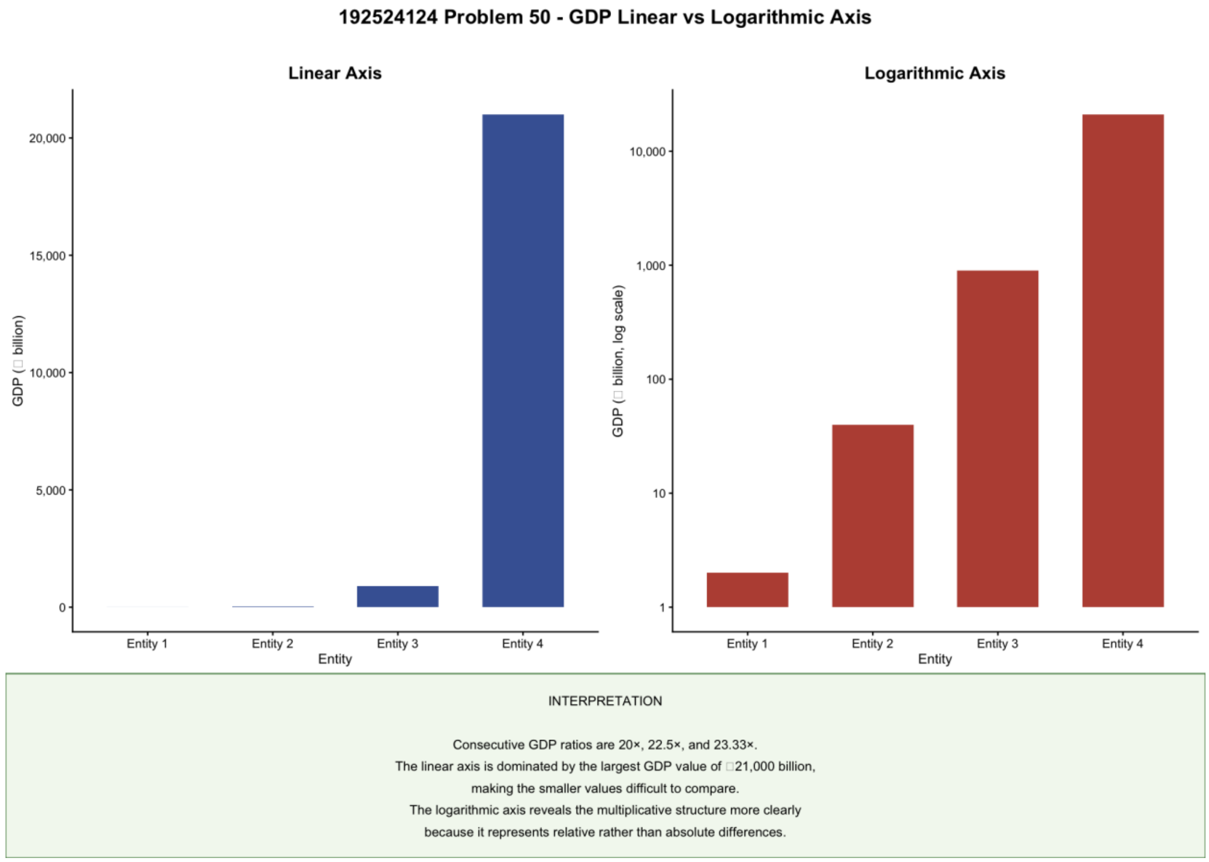

Figure 50: National GDP - Linear Axis (left) vs. Logarithmic Axis (right)